Monic quadratic trinomials

1 Factorise:

a $x^2 + 8x + 16$

b $x^2 + 16x + 64$

c $x^2 - 4x + 4$

d $x^2 - 20x + 100$

e $64 + 16x + x^2$

f $36 - 12x + x^2$

2 Complete the following statement: To factorise $x^2 + 9x + 18$, we need to find two numbers whose product is $\square$ and whose sum is $\square$.

3 Consider the quadratic $x^2 + 11x + 24$. To factorise this quadratic, we need to find two numbers.

a What should their product be?

b What should their sum be?

4 Given that $p < q$, find the values of p and q in the following pairs of equations:

a $p + q = 19$, $pq = 90$

b $p + q = -2$, $pq = -63$

c $p + q = -10$, $pq = 24$

5 Complete the following equations:

a $(m + 4)(\square) = m^2 + 10m + 24$

b $(y - 14)(\square) = y^2 - 12y - 28$

c $(m + 8)(\square) = m^2 + 18m + 80$

d $(x - 2)(\square) = x^2 - 6x + 8$

6 Find the values of m and n that will make the equation true: $y^2 + my + 35 = (y + 5)(y + n)$

7 Factorise:

a $x^2 + 6x + 8$

b $x^2 + 11x + 24$

c $x^2 + 17x + 72$

d $x^2 - 4x + 3$

e $x^2 - 17x + 60$

f $x^2 - 19x + 84$

g $x^2 - x - 6$

h $x^2 - x - 30$

i $x^2 + x - 20$

j $x^2 - 3x - 54$

k $x^2 - 3x - 70$

l $x^2 + 4x - 117$

m $40 + 13x + x^2$

n $35 - 12x + x^2$

o $-8 - 6x - x^2$

p $-12 + 7x - x^2$

8 Factorise:

a $x^4 - x^2 - 12$

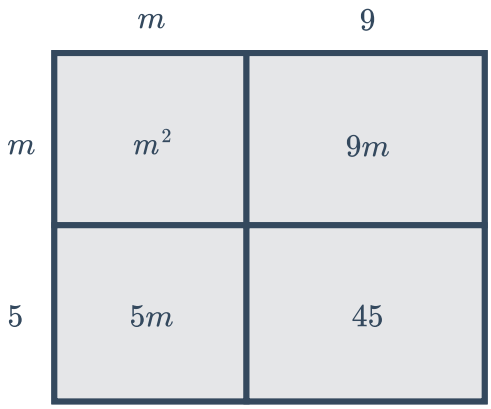
b $9y^4 - 13y^2 + 4$

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Applications

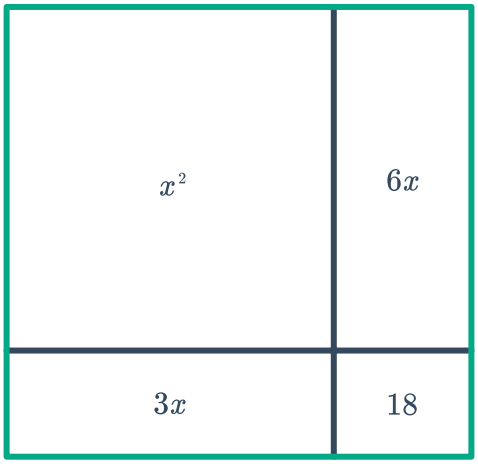
9 A square has an area of $x^2 + 6x + 9$. Find an expression for the length of a side. Assume that x is positive.

10 One expression for the area of the rectangle below is $m^2 + 14m + 45$. The rectangle is made up of four smaller rectangles. Use the diagram to express the area of the large rectangle in factorised form.



11 Find an expression for the total area of the following rectangles in factorised form:

a



b



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